MIL-T-60668(MU)
27 November 1968
SUPERSEDING
FA-PD-MI-2665
28 May 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M18
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated: M18, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F8597515	- Thruster, Cartridge Actuated: M18, Metal Parts Assembly
C8848865	- Container, Ammunition, Fiber M365 for Thruster, Cartridge Actuated: M18, with Cartridge M43
C8848864	- Box, Packing, Ammunition for Thruster, Cartridge Actuated: M18 with Cartridge M43 in Fiber Container M365
IEL-8597515	- Inspection Equipment Lists for Thruster, Cartridge Actuated: M18, Metal Parts Assembly

FSC 1377

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(Copies of specification, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the Ammerican Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Material.- Components for remanufacture shall be obtained from the M18 Thruster model. All components from earlier model types shall be modified in accordance with the latest component drawing. All components from earlier model types shall be identified and segregated for required modification prior to reassembly(See 6.4.3).

3.3 Hydrostatic pressure.- The Body and Cap components shall withstand a constant internal pressure of 18,000 pounds per square inch, (psi), minimum. The Head component shall withstand a constant internal pressure of 9,500 psi, minimum. The constant internal pressures shall be held for 15 seconds, minimum, without application of any external load. The components shall show no evidence of leakage or mechanical failure(See 6.2).

3.4 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal part assembly drawing, F8597515. All threaded joints shall be assembled with sufficient torque to lightly bottom components, except as noted in Table I.

3.4.1 The metal parts assembly shall be assembled with the following new metal part components: Shear Pin, 8597498; Dyna Seal Washer, 10534356; O-Rings; Silicone Fluid; Lubricants; and Special Compounds.

3.4.2 The metal parts assembly shall include component parts that comply with the following drawing requirements: The thread lead(.25 + .01) of each Body Assembly shall conform with drawing F8597516. The boss diameter(1.022 - .005) of each Head component shall conform with drawing F85974503. The midsection diameter(1.135 + .002) of each Cap component shall conform with drawing D8597500. Any components that do not comply with drawing requirements shall be either modified or discarded. Set screws or any thread component which shows evidence of cross threading, stripping or excessive wear shall be replaced.

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3.4.3 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I

Component	Seating or Bottoming (inch pounds)	Breakaway (inch pounds)
Body and Cap	600 ± 20	400 min.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied. (See 6.4.1)

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches. (See 6.3)

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance Terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 and as specified herein.

4.2 Inspection Provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Products", provisions of MIL-STD-105.

4.2.1.2 Size of Lot.- The shipping lots of metal parts assemblies shall consist of 1200 thruster, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimans prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed Parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be either reprocessed or rejected.

4.2.2.2 Reprocessed Parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 -22,000	300	4	5	5	6

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TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects Number of Defects in class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Sizes for Minor Defects Number of Defects in class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.2.2.3 Cap, Drawing B8597500 covering a detail of Drawing F8597515.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.4 Disc, Shear, Drawing A8597501 covering a detail of Drawing F8597515.

CRITICAL

None Defined.

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4.2.2.4 (Continued)

Defect	Method of Inspection
MAJOR	
101. Metal defective, creased or bent	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.5 Disc, Shear (Front), Drawing B8597502 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Metal defective, creased or bent	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.6 Head, Drawing C8597503 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

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4.2.2.7 Pin, Firing, Drawing B8597504 covering a detail of Drawing F8597515.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.8 Piston, Drawing B8597506 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.9 Piston end, Drawing B8597508 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish, improper	Visual
103. Burr	Visual
104. Foreign matter	Visual
105. Pellet loose, missing or improperly inserted	Visual

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4.2.2.9 (Continued)

Defect	Method of Inspection

MINOR

None defined.

4.2.2.10 Ring, Drawing B8597510 covering a detail of Drawing F8597515.

CRITICAL

None defined.

MAJOR

101.	Metal defective	Visual
102.	Protective finish inadequate or defective	Visual
103.	Burr or sharp corner	Visual
104.	Foreign matter	Visual

MINOR

None defined.

4.2.2.11 Body, Drawing B8597516 covering a detail of drawing F8597515.

CRITICAL

None defined.

MAJOR

101.	Thread damaged or not full	Visual
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr or sharp corner	Visual
105.	Pellet missing, loose or improperly inserted	Visual
106.	Foreign matter	Visual

MINOR

None defined.

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4.2.2.12 Piston (Assembly), Drawing C8597518 covering a detail of Drawing F8597515.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Metal defective.	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp edges	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.13 Tube Assembly, Drawing C8597520 covering a detail of Drawing F8597515.

CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Foreign matter	Visual
104. Burr or sharp edges.	Visual
MINOR	
None defined.	

4.2.2.14 Assembly, Drawing F8597515(less cartridge) covering a detail of Drawing D8593275.

CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
3. Evidence of oil leakage	Visual

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4.2.2.14 (Continued)

Defect	Method of Inpsection
MAJOR	
101. Thread damaged	Visual
102. Metal defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
201. Shipping cap missing	Visual
202. Protective finish inadequate or defective	Visual

4.2.2.15 Container, Fiber, Drawing C8848865.

CRITICAL

None defined.

MAJOR

None defined.

MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.16 Box, Packing, Drawing C8848864.

CRITICAL

None defined.

MAJOR

None defined.

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4.2.2.16 (Continued)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.2.2.17 Process control.- Periodic unscheduled monitoring shall be conducted for the following process operations to determine compliance with the applicable requirements of the indicated drawing:

Amount of silicone fluid	F8597515
Coat contact faces of tube and disc with sealing compound	F8597514
Installation of cap screw and washer on piston	F8597514
Bottoming torque	Requirement 3.4.3

4.2.3 Inspection Testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Hydrostatic pressure.- Each body, head and cap shall be subjected to hydrostatic pressure test. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the component.

4.2.3.2 Breakaway torque.- Each body and cap sub-assembly shall be subjected to a breakaway torque. Failure of any sub-assembly to comply with the requirement of 3.4.3. shall be cause for rejection of the sub-assembly.

4.2.3.3 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements shall be cause for rejection of the lot.

4.2.3.4 Residual magnetism.- Each thruster shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirement of 3.6 shall be cause for rejection of the assembly.

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4.2.4 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL 8597515.

4.3 Test Methods and Procedures.

4.3.1 Hydrostatic pressure.- Prior to the application of protective finish the body, cap and head, individually, shall be mounted in a device and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.3. The testing device used shall internally seal the body, cap and head, as applicable, at each end without applying any axial load other than applied by the liquid pressure.

4.3.2 Breakaway torque.- A wrench, set at the torque requirement specified in 3.4.3 shall be used in conjunction with a fixture to determine the adequacy of the joint. The body shall be secured and the breakaway torque shall be applied to the cap.

4.3.3 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetism effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Packing and marking shall be in accordance with Drawings C8848864 and C8848865.

5.2 Data Cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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6.3 When remanufacturing and loading of assembly is performed by the same facility, the requirement of 3.6 is waived and the residual magnetism test is performed on the loaded assembly only.

6.4 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.4.1 Protective finish.- Remanufactured exterior components which include Head, Body and Cap shall be stripped of protective finish and replated in accordance with the applicable drawings and specifications.

6.4.2 Nylon pellets.- Bodies which contain nylon pellets and are to be replated, shall have the pellets removed and new pellets installed after replating in accordance with the applicable drawings.

6.4.3 Modification.

6.4.3.1 Body components from T8E1 thruster model shall be modified with a thread lead (.25 + .01) and nylon pellets inserted as shown on drawing F8597516.

6.4.3.2 Head components from T8E1 thruster model shall be modified with a boss diameter (1.022 - .005) as shown on drawing F85974503.

6.4.3.3 Cap components from T8E1 thruster model shall be modified with a midsection diameter (1.135 + .002) as shown on drawing D8597500.

6.5 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2665 dated 28 May 1965.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A169

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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